Jai Mehta

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Education

University of Tennessee *Aug. 2023 – May 2027*
BS in Honors Computer Science **GPA: 4.00/4.00**
BA in Honors Computational Biology and Biomedical Machine Learning
Minors in Pre-Health Professions, Machine Learning

Wilson Central High School *May 2023*
Salutatorian, High School Diploma **GPA: 4.55/4.00**

Coursework

Computer Science Courses: Generative Protein Design, Introduction to Machine Learning, Bioinformatics, Honors Algorithms, Data Structures, Computer Organization, Discrete Mathematics, Algorithms Analysis and Automata, Honors Probability and Statistics, Linear Algebra, Honors Calculus I & II

Biology Courses: Cellular and Molecular Biology, General Chemistry (with labs), Organic Chemistry I & II (with labs), Biochemistry, Honors Physics I & II (Calculus-Based)

Experience

Visiting Undergraduate Researcher *Cambridge, MA*
Broad Institute of MIT and Harvard *Jan. 2026 - Present*

- Working for **Project Ex Vivo,** an interdisciplinary project between the Broad Institute, Microsoft Research, and Dana Farber Cancer Institute/Harvard Medical School.

Undergraduate Researcher *Oak Ridge, TN*
Oak Ridge National Lab, Center for Molecular Biophysics *Nov. 2023 - Present*

- Working on a pipeline using RFDiffusion, LigandMPNN, and Boltz-2x/AlphaFold3 to redesign enzymes for nylon waste breakdown, increasing activity up to 19-fold for various substrates.
- Developing an iterative optimization tool to improve enzyme-substrate affinity and target out-of-scope substrates.
- Utilized generative deep learning technologies to develop a *de novo* protein binder to inhibit cell division in *E. coli.*

Research Member *Knoxville, TN*
Emrich Lab *Aug. 2024 – Present*

- Employed XGBoost, SVM, LASSO, and Random Forest models to observe transcriptomic and genomic factors responsible for malaria resistance.
  - Used dimensionality-reduction techniques such as Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) to identify clusters against existing labels.
  - Harnessed Deep Neural Networks (DNN) for underlying marker discovery.
- Comparing protein structure modeling tools with protein-language models for intrinsically disordered proteins.

Teaching Assistant *Knoxville, TN*
Introductory Computer Science in C++ and Probability and Statistics *Aug. 2024 – Dec. 2025*

- For *ECE 313/317* - Probability and Statistics (and Honors course): *Aug. 2025 – Dec. 2025*
  - Hosted office hours and graded papers for a cohort of over 100 students.
  - Taught topics related to set theory, probability theory, Bayesian probability, Permutations/Combinations, Discrete and continuous variables, along with their distributions.

- For *COSC 102* - Introductory Computer Science in C++: *Aug. 2024 – Dec. 2025*
  - Hosted weekly office hours to assist students with labs for a cohort of over 200 students.
  - Assisted in topics related to learning Linux, basic data structures, dynamic memory, recursion, file I/O, searching algorithms, and binary logic.

Research Intern *Nashville, TN*
Lau Lab, Center for Computational Systems Biology, Vanderbilt University *Jun. 2025 - Aug. 2025*

- Developed an imaging solution platform for a 3D tissue map for aligned serial whole slide images of colorectal cancer.
- Applied foundation models and convolutional neural networks (CNN) to annotate patches and visualize embeddings.
- Used Visium spatial transcriptomics and scRNA-seq reference data for cell type deconvolution and association with foundation model embeddings for 2.5D morphology interpolation.

Research Intern *Osaka, Japan*
Systems Immunology Lab, Osaka University Immunology Frontier Research Center *Dec. 2024 - Jun. 2025*

- Building a Clonotype to Paratope converter using IMGT and MiXCR inputs that takes a V gene name, J gene name, and CDR3b sequence, and provides the complete amino acid sequence. [GitLab]
- Using MAFFT to number antibodies and adaptive immune repertoires with unprecedented accuracy and speed. Wrote sequence parsers to build insertions for sequences using compact maps in C++. [GitHub]

Research Intern *Memphis, TN*
Yu Lab, St. Jude Children's Research Hospital *July 2024 - Aug. 2024*

- Evaluated a multi-informational clustering algorithm using scATAC-seq data with scRNA-seq labels to observe misclassification levels between cell phenotypes. Utilized Euclidean and Cosine distance estimation methods.
- Tested "Memory-like" and "Effector-like" cell states defined by a Gaussian Mixture Model using Joint Embedding, Differential Expression, and Gene Set Enrichment Analysis on clinical CAR T data.
- Combined clinical CAR T data, ran through Gaussian Mixture Model, and tested to observe similarities between individual clinical datasets and combined data.

Publications

**** Denotes co-first authorship.***

- Mannik, J., **Mehta, Jai M.**, Parks, J.M., Mannik, J. A de novo binder targeting FtsA oligomerization stops late divisome recruitment and reveals a hidden constriction mechanism in Escherichia coli. *Under review, Nature Communications.*
- Bourgery, C.*, **Mehta, Jai M.***, Demerdash, O.N., Cooper, C.J., Carper, D.L., Zhou, M., Dishner, I.T., Cahill, J.F., Bocharova. V, Parks, J.M., Michener, J.K. Structural Determinants of Substrate Selectivity in a Family of Nylon Hydrolases. In preparation.
- Dishner, I.T., Bourgery, C., Carper, D.L., Cooper, C.J., Cahill, J.F., **Mehta, Jai M.**, Yakubov, S., Zhang, H.-H., Ogbonna, N.D., Damron, J.T., Sacci, R.L., Bocharova, V., Parks, J.M., Michener, J.K., Foster, J.C., 2026. Circularity in Sequence-Controlled Copolyamides Enabled by Regioselective Enzymatic Hydrolysis. *Journal of the American Chemical Society.*

Abstracts

- **Mehta, Jai M**., Yakubov, S., Bourgery, C., Cahill, J.F., Michener, J.K., Parks, J.M. Improving AI-based co-folding of enzyme-substrate complexes with conformational sampling, rescoring, and biochemically informed geometric filtering. Southeastern Regional Meeting of ACS, 2026.
- Le M., Evans J., Kaur H., Lee S., Pujara V., Simmons A.J, Kang S., **Mehta Jai M.**, Vaickus L., Lau K.S, Levy J. Selecting Informative Histologic Sections for Cost-Efficient 3D Spatial Transcriptomics and Tumor Microenvironment Reconstruction. [abstract]. In: Proceedings of the American Association for Cancer Research Annual Meeting 2026; Part 1 (Regular Abstracts); 2026 Apr 17-22; San Diego, CA. Philadelphia (PA): AACR; Cancer Res 2026;86(7 Suppl):Abstract nr 1470.

Presentations and Talks

Generative Deep Learning-based Design of a *De Novo* Protein Binder that Inhibits Cell Division in *Escherichia coli*

- UTK Undergraduate Research and Fellowships Discovery Day, Knoxville, TN, Sep. 2026.
- SASE National Conference, Computational Sciences Division, Pittsburgh, PA, Oct. 2026.

Redesigning Protein-Ligand Complexes Using Deep Learning

- UTK Undergraduate Research and Fellowships Discovery Day, Knoxville, TN, Sep. 2025.
- UTK BCMB Department Board of Advisors Presentation, Knoxville, TN, Dec. 2025.

A Deep Learning-based Pipeline for 3D Visualization and Annotation of Colorectal Tumor Tissues

- UTK Summer Research Scholars, Knoxville, TN, Jul. 2025.
- Vanderbilt Summer Scholars Symposium, Nashville, TN, Jul. 2025.
- UTK Strengths Day, Knoxville, TN, Sep. 2025.
- SASE National Conference, Computational Sciences Division, Pittsburgh, PA, Oct. 2025.
- **Gulf Coast Undergraduate Research Symposium**, Bioengineering Division, Houston, TX, Oct. 2026.
  - Won **BIOE GCURS 2026 Outstanding Presentation Award**

Uncovering Genomic and Transcriptomic Markers for Drug-resistant Malaria using Machine Learning

- Presentation Day, UTK Algorithms, Knoxville, TN, May 2025.

Defining Effector and Memory-like Cell State In in-vivo CAR T Cells from Multi-center Clinical Trials

- St. Jude Intern Presentation Day, Memphis, TN, Aug. 2024.
- UTK Undergraduate Research and Fellowships Discovery Day, Knoxville, TN, Sep. 2024.
- SASE National Conference, Computational Sciences Division, Boston, MA, Oct. 2024.
- Posters at the Nashville Capitol, Nashville, TN, Apr. 2025
- National Conference for Undergraduate Research, Pittsburgh, PA, Apr. 2025
- Exhibition of Undergraduate Research and Creative Achievement (EURēCA), Knoxville, TN, Apr. 2025.
  - Won **achievement award** for presentation out of 500 submissions.

Evaluating a Multi-Informational Clustering Algorithm using scATAC-seq and scRNA-seq

- St. Jude Intern Presentation Day, Memphis, TN, Aug. 2024.

Panelist for *Student Panel for Undergraduate Research, Knoxville, TN, Nov. 2024.*

Grants

Richard and Ann Ray Scholarship Grant	2026
Alvin and Jane Ann Nielsen Memorial Scholarship Grant	2026
University of Tennessee Office of Research Grant	2025
University of Tennessee AURA Grant	2025
University of Tennessee College Scholars Honors Grant	2025
University of Tennessee Honors Grant	2025
University of Tennessee Student Success Grant	2025
University of Tennessee Honors Grant Recipient	2024

- Awarded small grant to pursue summer research endeavors.

Selected Awards

ORNL Directorate Award for Best Paper, Biosciences.	2026
Goldwater Scholarship Recipient	2026
College Scholars Honors Program	2025

James D. Howard Scholar	2025
Tau Beta Pi Honors	2024
Cook Grand Challenge Engineering Honors	2023
Chancellor's University Honors	2023
National Merit Scholar	2023
Distinguished Tennessean Scholar	2023
Ned McWherter's Scholar	2023
National Youth Science Camp Attendee	2023
Eagle Scout	2023

Mentorship

Yasmin Shaker, High School Student, ORNL Center for Molecular Biophysics	2025

Technologies

Programming Languages: Python, C++, R, Assembly, Java, Bash

Skills: SciKit-Learn, ScanPy, Seurat, MacOS, Windows, Linux, Raspberry Pi, Arduino, Vim, Jupyter Notebook, GitHub, LaTeX, Single Cell Analysis, Machine Learning, High Performance Computing, Google Colab, AlphaFold, PyTorch.